Scoring the proboscis extension response

A step-by-step guide to labeling honey bee behavior from video, one harnessed bee at a time, and turning those labels into PER data you can analyze.

You will learn to

- Build your own ethogram: behaviors, keys and clear definitions
- Recognize a full proboscis extension and the behaviors around it
- Use slow motion and frame stepping to time each behavior
- Label behavior bouts with start and stop key presses
- Check your labels and fix mistakes
- Read each bee's PER result and export your data

You will need

- A laptop with Chrome, Edge, Firefox or Safari
- PER Scorer, at `kyobisk.github.io/per-scorer`
- Videos of harnessed bees (MP4 works best)
- Your lab's bee IDs, holder numbers and group list
- About 45 minutes for your first run-through, after watching the 2-minute walkthrough on the same site

1. What PER is and why we score it
2. Building your ethogram
3. A tour of the screen
4. Scoring a bee, step by step
5. How a bee gets scored
6. Saving and exporting your data
7. Comparing groups and transitions
8. Good scoring practice and troubleshooting
9. Check your understanding
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Screenshots show a sample project with two bees and a student-built ethogram.

PART 1

What PER is and why we score it

When a hungry honey bee's antennae touch sugar water, she reflexively extends her proboscis (the tongue-like mouthparts) to drink. This is the **proboscis extension response**, or PER. Because the response is easy to see in a bee held still in a small holder, it has become one of the standard ways to study learning, memory and sensory perception in insects.

In a PER experiment you present a stimulus to a harnessed bee, such as sucrose on the antennae or an odor she has learned to associate with food, and record whether and how quickly she extends her proboscis. You will score these responses from video, so that every call can be replayed, checked and timed to a single frame.

Words you will see

TERM	MEANING IN THIS TUTORIAL
Holder (harness)	The small tube or mount that holds the bee still with her head free. Each holder has a number.
Trial	One recording of one bee. PER Scorer treats each bee's whole video as one trial.
Latency	Time from 0:00, the start of the video, to the start of the first full extension, in seconds.
Bout	One continuous episode of a behavior, with a start time and a stop time.
Ethogram	The fixed list of behaviors you score, each with a written definition.

Why label by hand? A yes/no PER score for each bee is the main result, but recording when each behavior starts and stops also gives you response latency, response duration, and what the bee was doing instead of responding (grooming, struggling). PER Scorer turns your key presses into all of these numbers at once.

PART 2

Building your ethogram

PER Scorer starts with no behaviors. Before you score, you decide **what** you will record and write a definition for each behavior: what you must see for it to start, and what ends it. That list is your **ethogram**. Anyone reading your definitions should be able to score a video and get the same answer you did.

Writing good definitions

- **Describe what you see, not what you think it means.** "Proboscis crosses the line between the open mandible tips" can be checked on a frame; "the bee is hungry" cannot.
- **Give a start and a stop.** If you cannot say when a behavior ends, record it as a single *point* instead.
- **Keep the list short.** Five to eight behaviors is plenty. Each extra key is one more thing to watch for.
- **Decide which behavior is the response.** Exactly one behavior, usually the full proboscis extension, should be marked *Counts as PER* (Part 4, step 3).
- **Agree as a group** if you are pooling data, and use the same names. Your instructor can share one ethogram file with the class.

Example: an ethogram for a PER experiment

This is one reasonable ethogram, with suggested keys. Use it as a model for your own wording.

KEY	BEHAVIOR	TYPE	START WHEN… / STOP WHEN…
P	Full proboscis extension	Bout	Start on the first frame the proboscis extends past an imaginary line between the tips of the open mandibles. Stop when it retracts back behind that line.
E	Partial extension	Bout	The proboscis moves or unfolds but does not cross the mandible line. Start at first movement, stop when it is folded away again.
A	Antennal movement	Bout	Both antennae point toward the stimulus or sweep back and forth rapidly. Stop when they return to a resting position.
M	Mandible opening	Bout	The mandibles open with no proboscis extension. Stop when they close.
G	Grooming	Bout	The bee wipes her antennae or head with her forelegs (only possible if her legs are free in your holder). Stop when the wiping stops for more than one second.
S	Struggling	Bout	Vigorous attempts to escape: strong body, leg or abdomen movements against the holder. Stop when the bee is still again.
X	Sting extension	Bout	The sting shaft is visibly pushed out of the abdomen tip. Stop when it is withdrawn.
D	Defecation	Point	One key press at the moment a droplet appears. Point events have no duration.

Full versus partial. This is the call scorers disagree on most. If you have to replay a moment three times to decide, slow it to 0.25× and step frame by frame with `,` and `.`. If it still is not clear, it does not cross the line: score it as partial and add a note in your lab notebook.

Behaviors can overlap. A bee can extend her proboscis and move her antennae at the same time; just start both. Each behavior has its own key and its own row on the timeline.

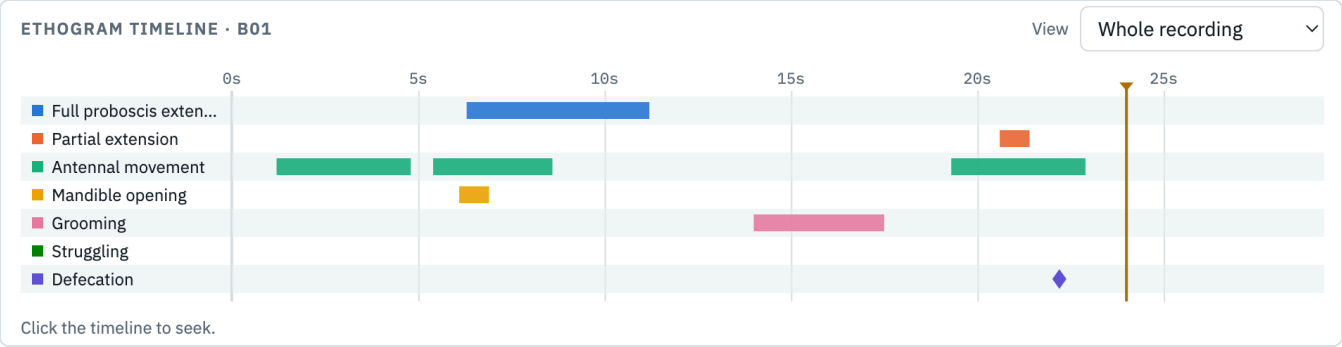
PART 3

A tour of the screen

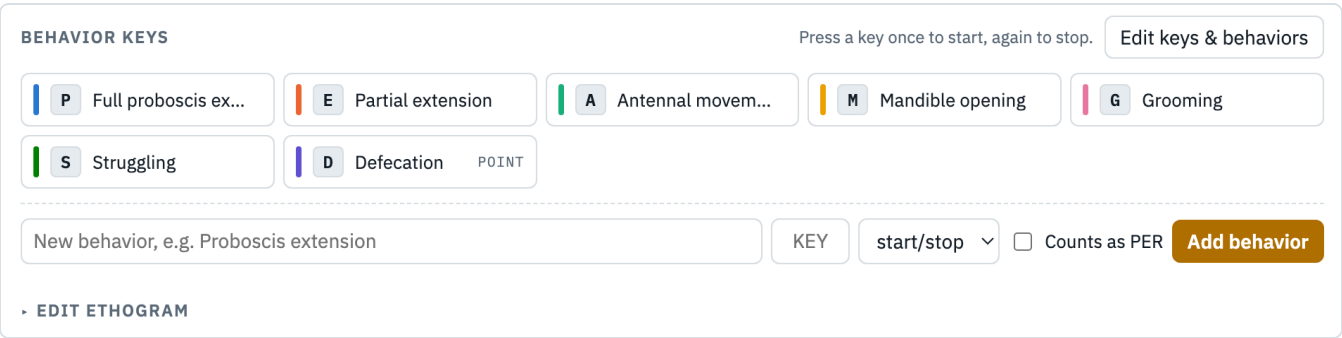
PER Scorer runs in a web browser. Nothing is installed and nothing is sent anywhere; your labels stay on your computer until you export them. The screen has a work area on the left and information panels on the right.



Video area and playback controls. Above the video: *Open video file...*, or a box for pasting a link to a video file. The badge in the top-left corner always names the bee you are scoring. Below the video: play, jump back or forward 1 or 5 seconds, step one frame, playback speed, frame rate, and the current time. With no video loaded, `Space` runs a stopwatch for live scoring.



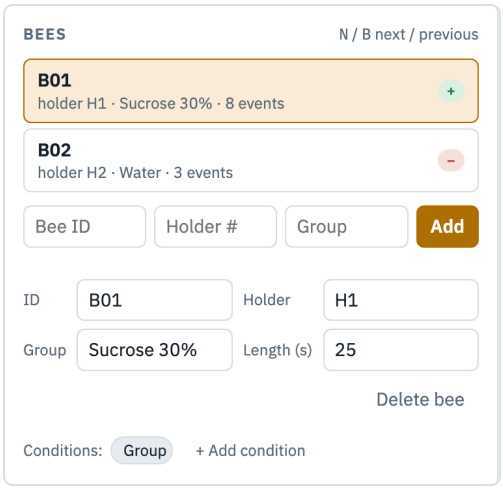
Ethogram timeline. One row per behavior. Bars are bouts, diamonds are point events, and the amber line is the playhead (the current time). Click anywhere on the timeline to jump the video there.



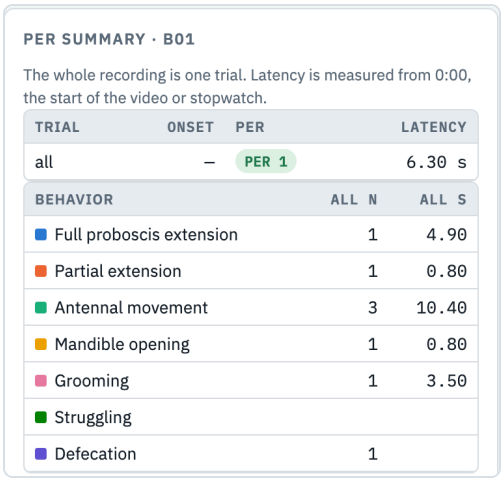
Behavior keys. The behaviors you have created, each with its key. A button turns amber while its bout is running and shows the elapsed time; you can click the buttons instead of pressing keys. Below them is the form for adding a behavior. *Edit keys & behaviors* opens the editor for renaming, recoloring or changing keys.



Event log. Every bout you have labeled for the current bee, newest first. You can type in a start or stop time to correct it, jump to the event (↦), or delete it (×). A bout that is still running is highlighted and has a *Stop now* button.



Bees. Your roster. The green + and red − marks show each bee's PER result; unlabeled bees say *not scored*. Add bees with the form. Below it you can edit the selected bee, including its recording length and any conditions your instructor asks you to fill in.



PER summary. For the selected bee: the PER result and latency, then the number of bouts (n) and total seconds (s) of each behavior over the whole recording.

PART 4

Scoring a bee, step by step

1 Open PER Scorer and fill in the session details

Open PER Scorer at `kyobisk.github.io/per-scorer`. At the top right, type your initials, check the date, and name the experiment (for example *Sucrose responsiveness, day 1*). These go into every row of your exported files.

2 Add your bees

In the **Bees** panel, type the bee's ID, holder number and treatment group, then click *Add*. The new bee becomes the selected bee. If the holder number ends in a digit, the form fills in the next one for you (H1, then H2…), so adding a whole rack is quick.

> **Group labels and blinding.** If your project is run blind, your instructor will give you coded group labels. Do not try to work out which group is which while scoring.

3 Create your behaviors

You only do this once per project. In the keys panel, type the name of a behavior from your ethogram, choose a key (PER Scorer suggests one from the name), choose *start/stop* or *point*, and click *Add behavior*. For the behavior that means the bee responded, tick **Counts as PER** before adding it. Repeat for each behavior. Each one gets a color, a button and a row on the timeline.

Keys must be single letters or digits, and each key can only be used once. `Z`, `N` and `B` are reserved for undo and switching bees. If your instructor gives you an ethogram file, skip the typing: click *Edit keys & behaviors* and

use *Import ethogram…*.

4 Load the video, or set up for live scoring

From video (recommended): click *Open video file…*, drag the file onto the black area, or paste a link to the video into the box next to the button and click *Load link*. A link has to point straight at a video file (usually ending in .mp4); YouTube links and Google Drive or Dropbox sharing links open a web page instead, so download those videos and open them as files. Set the **fps** box to the camera's frame rate (ask your instructor; 30 and 60 are common) so the frame-step buttons move exactly one frame.

Live, at the bench: skip the video. Press `Space` to start the stopwatch at the moment you begin, and score in real time. Live scoring is harder because you cannot rewind, so use it only if your instructor asks you to.

The scorer reads each bee's **recording length** from the video. You can see and correct it in the *Length (s)* box under the bee list; it is what proportions of time are calculated from.

Crop if the bee is small in the frame: click *Crop…* under the video, drag a box around the bee and its holder, and click *Apply* (or press `Enter`). That area then fills the video panel. *Show full frame* undoes it. Each bee keeps its own crop, and a new bee starts with the crop of the bee before it, because holders usually sit in the same place.

Trim if the recording has extra footage before or after the part you score: play to the first moment that counts and click *Start here*, then to the last moment and click *End here*. The start becomes 0:00 for that bee, so latency is measured from it, and the recording length is the trimmed length. Labels outside the trim are greyed out and not counted. Click a trim time to jump to it, or *Clear trim* to remove it. Neither crop nor trim changes the video file.

If several holders appear in the same video, score it once per bee: select a bee, crop to that bee's holder, score the video for that bee, then select the next bee and do the same.

> **Changing a behavior later.** Click *Edit keys & behaviors* to rename a behavior, change its key or color, or remove one you have not used yet. Renaming does not lose any labels. If your instructor has locked the names, you can still change keys and colors. *Export ethogram* saves your behaviors to a file you can reuse or share.

5 Label the bee's behavior

Make sure the video is at 0:00 (click the left end of the timeline) and press `Space` to play. Press a behavior's key once when it starts and once more when it stops. Watch the head and mouthparts closely, especially in the seconds after the stimulus is presented. If a lot happens at once, watch the clip through once first, then score it on a second pass.

- **Slow down** for the seconds around the stimulus: choose 0.5× or 0.25× in *Speed*, or press `[`.
- **Pause** with `Space` and step frame by frame with `,` and `.` to find the exact first frame of an extension. Then press the key while paused; the time is taken from the paused frame.
- **Mistake?** Press `Z` to undo the last change. You can undo many steps in a row.
- **Zoom** the timeline with *View* (8, 20 or 60 seconds around the playhead) when bouts are short.

6 Check your labels in the event log

Before moving on, read through the event log. Look for bouts still marked *open* (you forgot the stop press), and very short bouts from accidental key presses. Click ↦ to jump to any event and check it against the video. Fix times by typing new values.

7 Read the result and move to the next bee

The **PER summary** panel now shows PER 1 or PER 0 for the bee, with its latency. If a result looks surprising, go back and look at the video again before you change anything. Press `N` to select the next bee (or `B` to go back), and repeat steps 4 to 7.

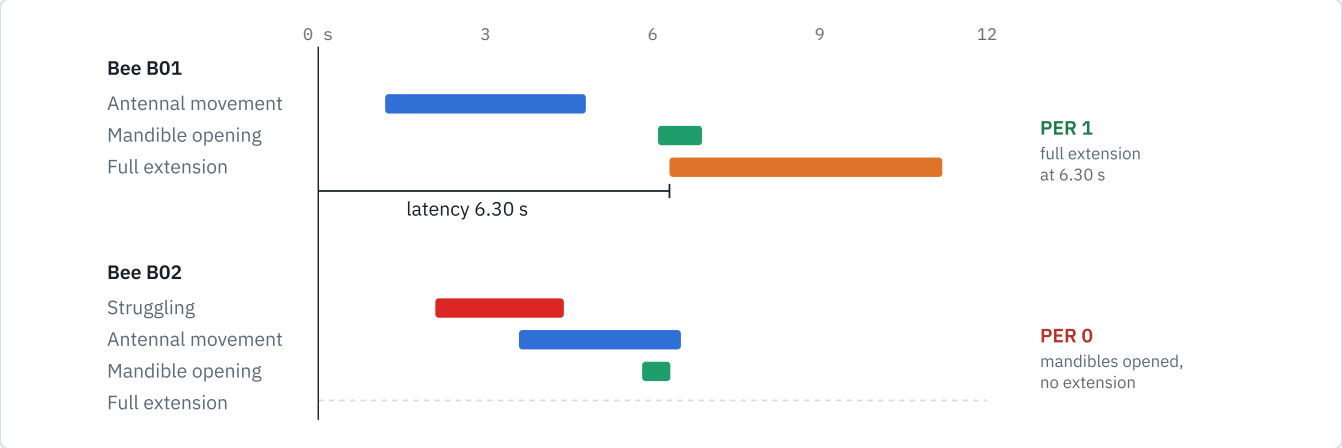
PART 5

How a bee gets scored

You never type PER 1 or PER 0 yourself. PER Scorer works it out from your labels with one rule:

> **The rule.** A bee scores `PER 1` if at least one bout of the behavior you marked **Counts as PER** appears anywhere in her recording. Otherwise she scores `PER 0`. **Latency** is the time from 0:00 to the start of the first such bout. In the examples below, that behavior is the full proboscis extension. Until a behavior is marked, PER shows as *not set*.

The diagram below shows the first 12 seconds of two sample bees, scored with the example ethogram from Part 2.



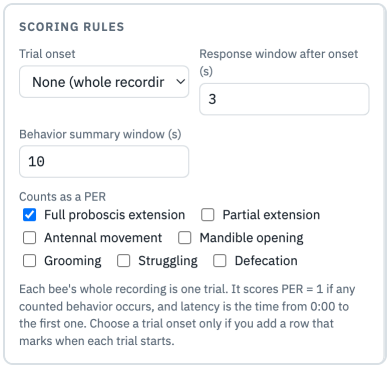
Bee B02 moved her antennae and briefly opened her mandibles, but her proboscis never crossed the mandible line, so she scores PER 0. Those other behaviors are still recorded, and they show up in the summary and the exported files.

> **Where is 0:00?** Because latency is measured from the start of the video, every video must start at the same moment relative to the stimulus, for example exactly when the sucrose touches the antenna. Your instructor will tell you how your lab's videos are trimmed. If a video starts at a different point, tell your instructor before scoring it.

Other things the summary reports

- **Latency**: seconds from 0:00 to the start of the first full extension. Blank when PER is 0.
- **n and s per behavior**: how many bouts started, and how many seconds were spent in each behavior, over the whole recording.

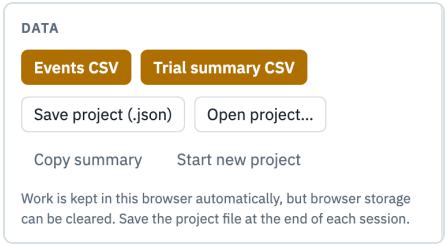
Scoring rules panel



Your instructor sets which behaviors count as a PER (for example, whether partial extensions count too). For experiments with several trials in one video, they can also choose a behavior that marks the start of each trial. **Do not change these on your own.** If two students use different rules, their results cannot be compared.

PART 6

Saving and exporting your data



PER Scorer saves your work as you go, in the browser you are using. Close the page whenever you like: when you open it again on the same computer, it returns to the same bee, reopens the video and goes to the moment where you stopped. The time of the last save is shown at the top of the page. Browser storage can still be wiped by clearing history or using a private window, and it does not move to another computer, so **save a project file at the end of every session** and keep it in your lab's shared folder.

- **Save project (.json)**: everything, including bees, labels and settings. Use *Open project...* to pick up where you left off, on any computer.
- **Events CSV**: one row per bout (bee, behavior, start, stop, duration, video file), for detailed analysis.
- **Trial summary CSV**: one row per bee, ready for a spreadsheet, R or Python.
- **Time budget CSV** and **Transitions CSV** (in the Analysis panel): per-bee time in each behavior, and transition counts, ready for R or Python.
- **Copy summary**: copies the trial summary so you can paste it straight into a spreadsheet, if saving files is blocked on your computer.

The all-bees table

ALL BEES · TRIAL RESULTS — 2 bees · 11 events

BEE	HOLDER	GROUP	T1	PER RATE	MEAN LATENCY
B01	H1	Sucrose 30%	1	100%	6.30 s
B02	H2	Water	0	0%	–
% responding			50%		

At the bottom of the page: every bee's result and latency, and the percentage of bees that responded.

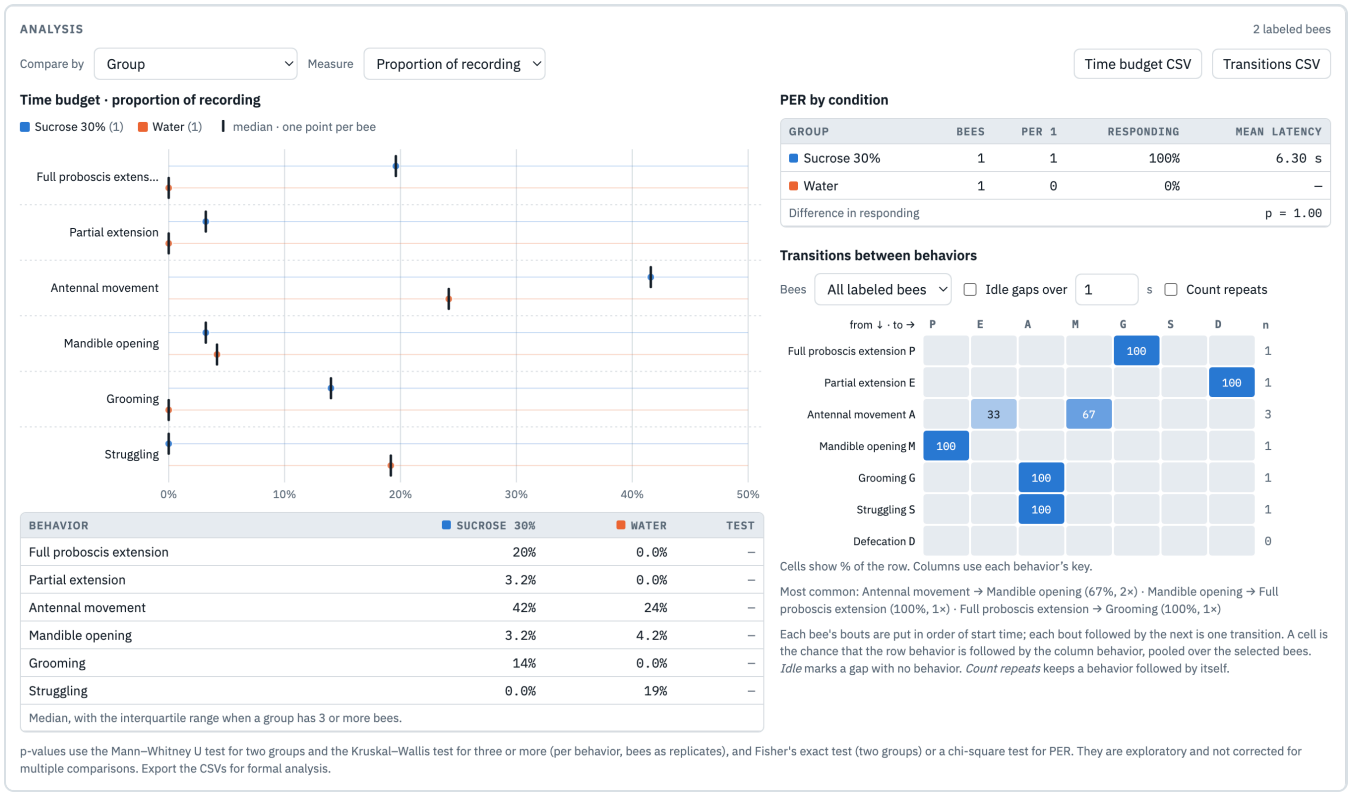
What is in the trial summary file

COLUMN	MEANING
`bee_id`, `holder`, `group`	From the Bees panel.
`trial`	*all*, meaning the whole recording.
`trial_onset_s`	Blank unless your instructor sets a trial-onset behavior.
`per`	1 = responded, 0 = did not.
`latency_s`	Seconds from 0:00 to the first full extension (blank when `per` is 0).
`pre_extended`	Only used when a trial-onset behavior is set; otherwise 0.
`response_window_s`, `summary_window_s`	Settings used for trial-onset scoring; not used for whole-recording scoring.
`<behavior>_count`, `<behavior>_dur_s`	Bouts started and seconds spent in each behavior.

PART 7

Comparing groups and transitions

The **Analysis** panel at the bottom of the page compares bees across conditions. It updates as you label, and uses every bee that has at least one label.



Analysis. Left: the time budget. Each behavior has one lane per group, with a point for each bee and a dark line at the group's median; the table below gives the median and interquartile range. Right: PER by condition, and the transition matrix. Hover over a bar or cell for its exact value.

Conditions

Every bee already has a **Group**. To compare bees by something else, click *+ Add condition* under the bee list (for example *Colony* or *Day*) and fill it in for each bee. Choose what to compare with *Compare by*.

Time budget

The default measure is **proportion of recording**: the time a bee spent in a behavior divided by the length of her recording. That keeps bees with longer or shorter videos comparable. You can switch to total seconds, number of bouts, or mean bout length. Point behaviors such as defecation only appear under number of bouts.

Transitions

Each bee's bouts are put in order of start time, and each bout followed by the next counts as one transition. A cell in the matrix is the chance that the row behavior is followed by the column behavior. For example, 67% in the *Antennal movement* row under *M* means two of every three antennal bouts were followed by mandible opening. Tick *Idle gaps* to count pauses longer than a set time as their own state.

> **About the p-values.** The panel shows a quick test for each comparison (Mann–Whitney or Kruskal–Wallis for time budgets, Fisher's exact or chi-square for PER). They are exploratory: they treat bees as independent and are not corrected for testing many behaviors at once. Use the exported CSVs for your real analysis.

PART 8

Good scoring practice

- **Score from the definitions, not from memory.** Reread Part 2 at the start of every session.
- **Do two passes when it helps.** Watch a busy clip once at normal speed, then score it. It is much easier than trying to catch everything at once.
- **Score blind when you can.** Knowing a bee's treatment can nudge borderline calls, even without meaning to.
- **Check the sucrose response.** A bee that does not extend when sugar touches her antennae is not motivated (or not healthy), and her zeros say nothing about the stimulus you are testing. Note these bees; your lab will decide whether to exclude them.
- **Never edit a result to match what you expected.** Only change a label after looking at the video again.
- **Check agreement.** Your instructor may have two people score the same videos and compare results. Disagreement on more than the occasional trial means the definitions need discussing, not that one person is wrong.
- **Take breaks.** Attention drops after about 30 minutes of close video work, and missed short bouts are the first sign.
- **Keep a notebook.** Write down anything odd: a bee that escaped, a blurry stretch of video, a stimulus that was delivered late.

Troubleshooting

PROBLEM	WHAT TO DO
The video will not play	The browser cannot read that format. Ask your instructor for an MP4 (H.264) copy, or convert it with a free tool such as HandBrake.
PER shows "not set"	No behavior counts as a PER yet. Tick *Counts as a PER* for your response behavior under Scoring rules.
A video link will not load	The link has to point straight at a video file. YouTube links and Drive or Dropbox sharing links open a web page; download the video and use *Open video file...*.
I pressed a key and nothing happened	The cursor is probably in a text box. Click on an empty part of the page, then try again.
I forgot to press stop	Find the open bout in the event log and type the correct stop time, or jump to the right moment and click *Stop now*.
I labeled the wrong bee	Press `Z` repeatedly to undo, or delete the events in the log, then select the right bee and score again.
It asks me to open the video again	The video could not be kept in this browser (it may be very large, or you are on a different computer). Open the same file again; your labels are still there and it goes back to where you stopped.
A warning says the scorer is open in another tab	Close the extra tab or window. Two copies of the page save over each other.

My bees disappeared	The browser storage was cleared. Use *Open project…* with your last saved .json file. This is why you save at the end of every session.
The export button does nothing	Some computers block downloads from the page. Use *Copy summary* and paste into a spreadsheet, or ask for the `per-scorer.html` file and open it directly.

PART 9

Check your understanding

Answer these before you score real data. Assume the default settings: the whole recording is one trial, and only full extensions count as a PER.

1. A bee's proboscis first crosses the mandible line at 13.8 s. What are her PER score and latency?
2. The proboscis unfolds at 7.1 s, but never passes the tips of the mandibles. Using the example ethogram in Part 2, what do you label it as, and what is the bee's PER score if nothing else happens?
3. You finish a bee and the event log shows one Grooming bout as *open*. What happened, and how do you fix it?
4. A bee fully extends from 4.2 to 5.0 s and again from 9.0 to 12.5 s. What are her PER, latency, and the n and s for Full proboscis extension?
5. You scored 20 bees yesterday on a lab computer. Today the Bees panel is empty. What do you do?
6. Why does it matter that every video starts at the same moment relative to the stimulus?

Answers. 1. PER 1, latency 13.80 s. 2. Partial extension. PER 0, because only the behavior marked Counts as PER (full extension) counts. 3. You never pressed stop. Type the stop time in the log, or jump to the right moment and use *Stop now*. 4. PER 1, latency 4.20 s, n = 2, s = 4.30 (0.8 + 3.5). 5. Open your saved project file with *Open project…*. If you did not save one, the labels are gone and the videos need to be rescored. 6. Latency is measured from 0:00. If videos start at different points, latencies from different bees cannot be compared.

PART 10

Quick reference card

Print this page and keep it next to your keyboard.

Playback

`Space`	Play / pause (or start / stop the stopwatch)
`←` `→`	Back / forward 1 s
`Shift` + `←` `→`	Back / forward 5 s
`,` `.`	One frame back / forward
`[` `]`	Slower / faster
Click timeline	Jump to that time

Editing and bees

`Z`	Undo last change
`N` / `B`	Next / previous bee
Event log	Type new times, ↦ jump, × delete
Edit keys & behaviors	Change a key or color

My keys: press to start, press again to stop

☐ _____

☐ _____

☐ _____

☐ _____

☐ _____

☐ _____

☐ _____

☐ _____

Write in your own behaviors. Circle the one that counts as a PER.

Every bee

1. Select the bee (check the badge on the video)
2. Go to 0:00 (your behaviors are already set up)
3. Press `Space` to play
4. Label behaviors, slowing down near the stimulus
5. Check the event log for open or accidental bouts
6. Look at the PER summary, then press `N`

Every session: fill in initials, date and experiment first. Save project (.json) before you close the browser.